UNDER A COSIMPLICIAL RESOLUTION AND AN ISOMORPHISM OF L PI_! ON THE STRUCTURE SHEAVES, L PI_! K AGREES WITH L PI_! OF THE EXTENDED SCALARS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to bounded above complexes and to the flat sheaves j_{U!}O_U, the definition of the simplicial abelian groups X_bullet and X'_bullet, the functoriality argument giving the homotopy equivalences with P_bullet and P'_bullet, and the conclusion from the assumed quasi-isomorphism.

Editorial note. The label lemma-O-homology-qis names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The functor L pi_! (homology of a site) and its computation by a cosimplicial resolution. Same chapter, the left derived functor of pi_! and the lemma computing it on j_{U!}O_U through the morphism complex of a cosimplicial object U_bullet. Supplies the identification of the complex X_bullet with L pi_!(j_{U!}O_U) and of P_bullet, P'_bullet with L pi_! O, L pi_! O'.
- The chaotic topology and the flatness of j_{U!}O_U. Same chapter, the structure of sheaves on a category with chaotic topology: every bounded above complex of O-modules is quasi-isomorphic to one whose terms are direct sums of j_{U!}O_U. Supplies the reduction of the computation to the single sheaves j_{U!}O_U, together with their flatness, which makes tensor with O' exact.

Source reference. Original proposition 0794: sources/stacks-chow/tags.txt, source lines 9259–9259. Prerequisite pointer invoked at proof line(s) 10444; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08Q9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10229–10229. Prerequisite pointer invoked at proof

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line(s) 10428; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EW: sources/stacks-chow/tags.txt, source lines 4471–4471. Prerequisite pointer invoked at proof line(s) 10449; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a category (endowed with chaotic topology). Let $\mathcal{O} \to \mathcal{O}'$ be a map of sheaves of rings on $\mathcal{C}$. Assume*

(1) *there exists a cosimplicial object $U_\bullet$ in $\mathcal{C}$ as in Lemma 08Q9, and*
(2) *$L\pi_!\mathcal{O} \to L\pi_!\mathcal{O}'$ is an isomorphism.*

For K in $D(\mathcal{O})$ we have

$$L\pi_!(K) = L\pi_!(K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}')$$

in $D(Ab)$.

Proof. Note: in this proof $L\pi_!$ denotes the left derived functor of $\pi_!$ on abelian sheaves. Since $L\pi_!$ commutes with colimits, it suffices to prove this for bounded above complexes of $\mathcal{O}$-modules (compare with argument of Derived Categories, Proposition 0794 or just stick to bounded above complexes). Every such complex is quasi-isomorphic to a bounded above complex whose terms are direct sums of $j_{U!}\mathcal{O}_U$ with $U \in \mathrm{Ob}(\mathcal{C})$, see Modules on Sites, Lemma 03EW. Thus it suffices to prove the lemma for $j_{U!}\mathcal{O}_U$. By assumption

$$S_\bullet = \mathrm{Mor}_{\mathcal{C}}(U_\bullet, U)$$

is a simplicial set homotopy equivalent to the constant simplicial set on a singleton. Set $P_n = \mathcal{O}(U_n)$ and $P'_n = \mathcal{O}'(U_n)$. Observe that the complex associated to the simplicial abelian group

$$X_\bullet : n \longmapsto \bigoplus\nolimits_{s \in S_n} P_n$$

computes $L\pi_!(j_{U!}\mathcal{O}_U)$ by Lemma 08Q9. Since $j_{U!}\mathcal{O}_U$ is a flat $\mathcal{O}$-module we have $j_{U!}\mathcal{O}_U \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}' = j_{U!}\mathcal{O}'_U$ and $L\pi_!$ of this is computed by the complex associated to the simplicial abelian group

$$X'_\bullet : n \longmapsto \bigoplus\nolimits_{s \in S_n} P'_n$$

As the rule which to a simplicial set $T_\bullet$ associated the simplicial abelian group with terms $\bigoplus_{t \in T_n} P_n$ is a functor, we see that $X_\bullet \to P_\bullet$ is a homotopy equivalence of simplicial abelian groups. Similarly, the rule which to a simplicial set $T_\bullet$ associates the simplicial abelian group with terms $\bigoplus_{t \in T_n} P'_n$ is a functor. Hence $X'_\bullet \to P'_\bullet$ is a homotopy equivalence of simplicial abelian groups. By assumption $P_\bullet \to P'_\bullet$ is a quasi-isomorphism (since $P_\bullet$, resp. $P'_\bullet$ computes $L\pi_!\mathcal{O}$, resp. $L\pi_!\mathcal{O}'$ by Lemma 08Q9). We conclude that $X_\bullet$ and $X'_\bullet$ are quasi-isomorphic as desired. □